

EMPLOYMENT PASSPORT

A Corporate Transparency Platform for Bangladesh's Labour Market

Whitepaper — Blockchain Olympiad Bangladesh 2026 · Category: Blockchain (Student)

Primary Domain: Identity & Privacy · Secondary: E-Governance · Tertiary: Fintech

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Platform: Hyperledger Fabric (permissioned) with daily Merkle-root anchoring to a public chain

Employment Passport — A Corporate Transparency Platform for Bangladesh's Labour Market

Draft whitepaper · Team 404_found_us · BCOLBD 2026 · (working draft — section by section)

1. Executive Summary

Bangladesh moves more than a million workers abroad each year and receives about \$30 billion in remittances, yet the systems that carry those workers — and the companies that employ them — run on paperwork that is easy to forge and impossible for a stranger to verify. The same failure repeats at every level: a worker cannot prove her skill or wage history to a foreign employer; a foreign employer cannot tell a certified electrician from a forged certificate; a company cannot prove that it pays its workers or who actually owns it; and an anonymous online review of an employer, however sincere, carries no more weight than gossip.

Employment Passport solves this by becoming a **corporate-transparency platform**. Every fact about a person, a workplace, and a company — a skill, a job, a wage, a contract, an ownership stake, a review — is issued as a **verifiable digital credential**, signed by whoever is responsible for it, and anchored to a shared ledger whose history no single party can rewrite. From those credentials the platform assembles five surfaces:

1. **Professional identity** — a LinkedIn-style profile in which every skill and job is *attested*, never self-typed.
1. **Workplace transparency** — a Deshi Mula-style employer review in which each review is provably written by a real employee, yet the author stays anonymous.
1. **Corporate identity** — a verifiable company registry, beneficial-ownership proof, and reconciled wage and procurement disclosure.
1. **Corporate Operations Suite** — a paid service that runs a workforce's lifecycle (onboarding, payroll, compliance, deployment logistics) so that *managing* workers is the same act as *producing* verifiable records.
1. **Labour-law enforcement** — a loop in which a worker's review that reports a violation automatically and provably reaches the ministry.

The reason this must be a blockchain, and not another database, is the central fact of the environment: ****institutions and paperwork in Bangladesh are not trusted, because they have repeatedly failed to be trustworthy.**** A conventional database has an owner, and whoever owns it can change it. A corrupt office, a dishonest employer, or even the platform itself can silently edit a record. Employment Passport removes that single point of trust: records are written only when the **required parties co-sign them** (an employer and a bank together sign a wage event), and the whole history is anchored daily to a public chain — so that not even the platform, nor the consortium behind it, can rewrite the past without being caught.

The result is a system whose promise is not "trust us," but *****"you do not have to trust anyone — verify the proof."***** For a worker, that means a credential that finally travels with her. For a foreign employer, it means hiring a Bangladeshi worker without trusting any Bangladeshi institution. For a company, it means a verified reputation that wins government tenders and foreign contracts. For the government, it means a labour market whose violations surface automatically. And for everyone, it means a transparency platform that is itself incapable of quietly inflating or suppressing a score.

1.1 What we are asking the jury to evaluate

- **The problem is multi-party and trust-broken** — five parties (worker, domestic employer/bank, foreign employer, regulator/public, and the platform itself) whose incentives and distrust no conventional system resolves.
- **Blockchain is necessary, not decorative** — §2.4 shows precisely what a conventional database cannot do here.
- **Privacy is not sacrificed to transparency** — selective disclosure and zero-knowledge proofs keep worker and company data out of the record while making the *claims* verifiable.
- **The architecture is specific and governed** — Hyperledger Fabric, six channels, per-schema endorsement policies, a nullifier registry, and daily public anchoring.

- **It is commercially real** — a paid Operations Suite funds the network, and the value it produces (verified trust) is exactly what foreign and government buyers pay for.

2. The Problem

2.1 Scale and stakes

The numbers define the stakes. Bangladesh sends more than a million workers overseas each year and receives about \$30 billion in remittances — among the highest-leverage variables in its development arithmetic. Domestically, roughly 20,000 computer-science and engineering graduates enter the job market each year against only 7,500–8,000 IT openings, forcing graduates to choose employers on marketing rather than evidence. Across both markets the same three assets are at stake: a worker's history, a company's identity, and a company's reputation as an employer. All three exist today only as paper or as self-asserted claims — and paper is forgeable, while self-assertion is free.

2.2 Concrete failure modes

The platform exists because each of the following is a real, recurring, and costly failure:

1. **Fake paperwork.** Certificates are photocopied and forged; a laminated card or a scanned PDF proves nothing about who issued it or whether it was ever earned.
1. **Contract substitution.** Employers swap terms after arrival because no one witnessed the original; the worker holds no copy that a third party will accept.
1. **Unverifiable reviews.** An anonymous employer review — however genuine — cannot be distinguished from a competitor's smear or a company's own astroturfing, so the market discounts it to zero.
1. **Foreign employers cannot verify.** A Gulf or European employer has no way to authenticate a Bangladeshi worker's skill, conduct, or wage history without trusting an institution it has no reason to trust, so it discounts every claim and depresses wages for everyone.
1. **Labour law cannot be enforced.** Wage theft, unpaid overtime, and unsafe conditions are rarely proven, because the payroll and the "record" both live in the employer's own database, which the investigator must take on faith.
1. **Corporate opacity.** Who actually owns a company, and whether its disclosed wage bill matches what it pays, are unknowable to the public and even to the government without an audit it cannot routinely perform.

2.3 Stakeholders and the five-party trust map

Each stakeholder needs something it cannot obtain on its own, and — crucially — none of them trusts any other enough to build it:

Stakeholder	What they need	Why they cannot fix it alone
Worker / migrant	Portable proof of skill, contract terms, wage history	Holds no institutional power; a self-asserted record is worthless
Recruiting agency / employer	Cheap screening; credible differentiation	Opacity is profitable for the worst actors; the industry cannot self-regulate
Destination / foreign employer	Confidence that a hire can do the job	Cannot audit foreign training centres or documents; defaults to discounting all claims
Government (MoEWOE/BMET, RJSC, DIFE)	Reduced fraud, enforceable labour law, verifiable companies	A national database is not authoritative abroad, and its own offices are not fully trusted
Bank / MFI / insurer	Verified income for credit and insurance	No trustworthy income signal for the informally documented
The platform itself	Must be prevented from becoming the new single point of failure	Even the operator cannot be trusted to keep a neutral score when a paying client is involved

The pattern is the point: the problem is not that the data does not exist — it is that ****no party is trusted to hold it, and every party's incentive is to manipulate it.**** This is a coordination-and- trust failure, not an information-storage failure — precisely the class of problem a shared, governed, publicly-anchored ledger is built for.

2.4 Why a conventional database does not solve this

A conventional database — PostgreSQL, or any central system — has exactly one disqualifying property: **it has an owner**. Whoever administers it can update any row, delete any record, and rewrite history, and no outsider can tell. In this environment that single fact is fatal:

What the platform needs	What a database gives	What the ledger gives instead
A record no single office can fabricate	The admin can insert any row	Multi-party endorsement — a wage event is valid only if employer <i>and</i> bank co-sign
A history no one can silently rewrite	The admin can UPDATE/DELETE	Append-only, hash-chained blocks — tampering is detectable, not prevented by trust
"Was this revoked before or after the verification?"	A mutable column, answerable only by the admin	An ordered, tamper-evident status list every verifier reads
Auditability by outsiders who hold no account	A read endpoint whose honesty must be assumed	A daily Merkle-root anchor to a public chain — verifiable by anyone
Trust between institutions that distrust each other	Each runs its own copy and disagrees	Joint governance with no single controlling party

The deeper reason is specific to Bangladesh: ****any system whose integrity depends on the honesty of one administrator inherits that administrator's corruption.**** The platform is built on the opposite assumption — that any single official, employer, agency, or even the platform operator may be dishonest — and is therefore designed so that no one of them can alter the record alone, and even all of them together cannot alter it silently. Blockchain does not make dishonest people honest; it makes dishonesty **permanent, attributable, and expensive**, and it lets a foreign employer, a worker, or a journalist verify a fact without ever asking an institution for permission or trusting its word.

3. The Solution: a corporate-transparency platform

3.1 Concept

Employment Passport assembles five surfaces on top of one identity, credential, and anchoring layer. Each surface answers a different question about the same underlying trust gap, and each is unfalsifiable in the same way: a claim is only as good as the *signature* behind it, and a history is only as good as the *anchor* that makes it tamper-evident.

Surface	Question it answers	Built from
Professional identity	"What can this person provably do?"	Skill, contract, and wage credentials
Workplace transparency	"What is it actually like to work here?"	Verified-anonymous employee reviews
Corporate identity	"Who is this company, and what does it owe?"	Registry, ownership, procurement, wage disclosure
Operations Suite	"How does a company run a compliant workforce?"	The paid workflow layer that produces the records
Labour-law enforcement	"Does a violation reach the authority that can act?"	Review signals routed to the ministry

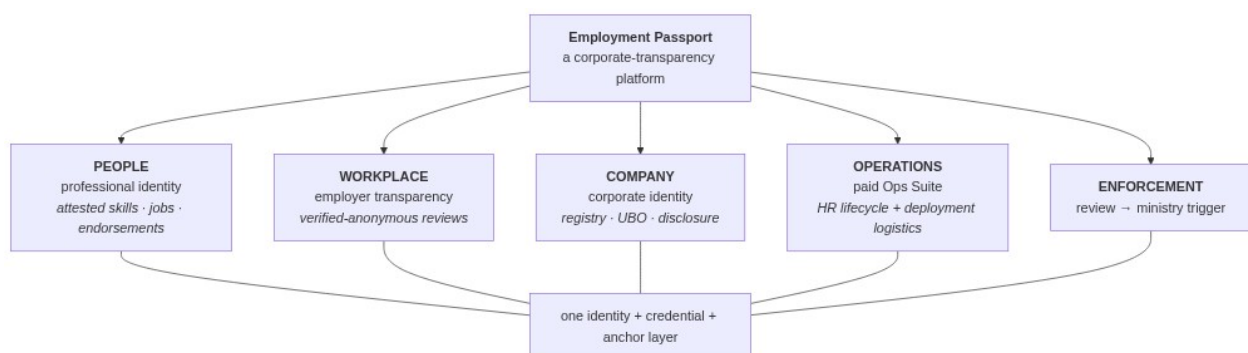


Figure 1 — The five surfaces of Employment Passport

The design principle throughout is that **no surface stores the sensitive data it verifies**. A review never stores the reviewer's identity; a disclosure never stores the full cap table; a wage event never stores the bank account. The ledger holds only salted hashes and proofs; the actual documents live in encrypted, off-chain

vaults. Transparency is therefore always **granular**: the world can verify a fact without being able to read the person or company behind it.

3.2 Credential types

Credentials are issued with **BBS+ signatures**, which allow a holder to prove an arbitrary subset of attributes — and predicates over them — without breaking the signature or revealing the rest.

Credential	Issuer	What it proves
SkillCertificate	TTC / trainer	A standardised skill and level
EmploymentContract	employer + agency	The agreed terms, signed at onboarding
EmigrationClearance	BMET	Legally cleared to work abroad
WageEvent	employer + bank	A specific, dated wage payment
EmploymentProof	employer	"This DID worked here, from A to B"
WorkplaceReview	worker (ZKP-gated)	A verified employee's assessment
Endorsement	a peer/manager (staked)	A specific, revocable claim of competence
CompanyRegistration	RJSC	Legal existence and standing
BeneficialOwnership	company + auditor + RJSC	Ownership thresholds, proven selectively
ProcurementAward	procuring entity	A tender outcome and conflict-check result

3.3 Selective disclosure in practice

Three worked examples show what "verifiable without revealing" means, beyond the identity layer:

- **A bank** reviewing a migrant-worker loan asks: *"paid at least twelve consecutive months at BDT 25,000+ in the past eighteen?"* *The worker returns a zero-knowledge proof that answers yes**; the bank learns nothing about the employer, the exact salary, or the country.
- **A foreign employer** asks: *"certified electrician, level 3+, no unresolved conduct finding?"* The proof does not reveal age, marital status, home district, or former employers' names.
- **A government tender** asks a bidder: *"no beneficial owner above 25% who is sanctioned or a politically exposed person?"** The bidder proves the threshold holds **without** disclosing the cap table — the government sees compliance, not the company's owners.

3.4 Professional identity (the LinkedIn layer)

A worker's profile is not a page of typed claims; it is a **wallet of issued credentials**. Skills, jobs, and wages arrive signed by the institutions responsible for them, so a profile is trustworthy in a way a self-authored résumé cannot be. Three properties distinguish it from a conventional professional network:

- **Attested endorsements.** A manager's "endorsement" is a staked, revocable credential, not a click. A false endorsement costs the endorser standing, so endorsements carry weight.
- **Corroborated reputation.** A claim supported by payroll *and* an independent bank record outranks a self-claim — the platform reports the *evidence basis*, not a bare score.
- **Privacy-preserving matching.** An employer searches "level-3 electrician, no conduct findings" and receives ZKP matches; the platform never sees the raw résumés it is matching.

3.5 Workplace transparency (the Deshi Mula layer)

Existing employer-review sites face a fatal weakness: ***anyone can post, and no one can tell a real employee from an astroturfing campaign.* **Employment Passport resolves this with verifiable anonymity**** — the reviewer is provably an employee, and provably anonymous, at the same time.

The mechanism is the **nullifier registry**. A reviewer proves, in zero knowledge, that she holds a valid, unrevoked `EmploymentProof` for the company she is reviewing — and submits a one-time value $n = \text{PRF}(\text{linkSecret}, \text{companyDID})$ that is unique to (this worker, this company) and reveals neither. Chaincode accepts the review only if n has never been used. The result is ***one verified voice per employee per employer*** — a company cannot stack its own ratings, and no single employee can post a hundred. Because each employer relationship uses a distinct link secret, a worker's reviews across different employers cannot be linked to each other or to her.

What the public sees is not the raw reviews but a **transparency digest**: "12 verified reviewers · 74% recommend · no open conduct findings" — each figure a commitment whose root sits in the anchored Merkle tree, so the aggregate can be verified without ever reading a single review.

3.6 Corporate identity and the Operations Suite

The corporate layer turns the company itself into a verifiable subject. Every firm holds a ****company DID**** backed by a regulator-verified `CompanyRegistration`; ownership is proven through **beneficial-ownership threshold proofs** rather than a published cap table; procurement awards are anchored with their conflict-check results; and a company's disclosed wage bill can be reconciled against the same anchored wage events that power the reviews.

The **Corporate Operations Suite** is the paid application that makes corporate participation a product rather than an obligation. It manages the workforce lifecycle — onboarding and bulk credential issuance, payroll with wage-event anchoring, clearance and visa tracking, deployment logistics, benefits, and offboarding — so that *running* a compliant workforce is the same act as *producing* the verifiable records that build trust.

The commercial logic is deliberately tight: a corporate customer pays because the Suite produces the **trust signal that foreign employers and the government both rely on**, which raises its value in winning government tenders, foreign job orders, and subcontracts. But the Suite is an ****unprivileged client**** of the same chaincode under the same endorsement policies as everyone else — a customer can buy the *tooling*, never the *score*. Its reputation is earned only from anchored, multi-party-signed evidence, which is precisely why the trust signal is worth anything.

3.7 Labour-law enforcement (the review → ministry loop)

Reviews are not only reputation; they are the **earliest available sensor for labour-law violations**. A worker knows she was not paid before any inspector does, and the platform routes that signal to the authority that can act.

- Each review carries a **violation taxonomy** — **wage paid on time? contract honoured? overtime paid? conditions safe? termination dues settled?** A "no" is a violation flag, not a sentiment.
- The flag is **verified and corroborated** before it escalates: the reviewer holds a valid employment proof, and the signal is either matched against anchored wage events or confirmed by independent reviews above a threshold.
- When the threshold is met, chaincode emits a `ViolationSignal(companyDID, code, evidenceHash)` and the platform routes a **scoped alert to the ministry** (MoEWOE/DIFE) — with the evidence, ****without the reviewer's identity and without any free text****.

The ministry investigates under its own legal authority and anchors an `InspectionReport` outcome, which updates the company's transparency score. The loop therefore closes: a worker reports, the ministry acts, and the outcome is itself verifiable. Two guardrails keep it legitimate — ****the platform supplies evidence, it never adjudicates, and selective disclosure prevents surveillance**** (the ministry receives only the facts the specific case requires, with legal authority or worker consent).

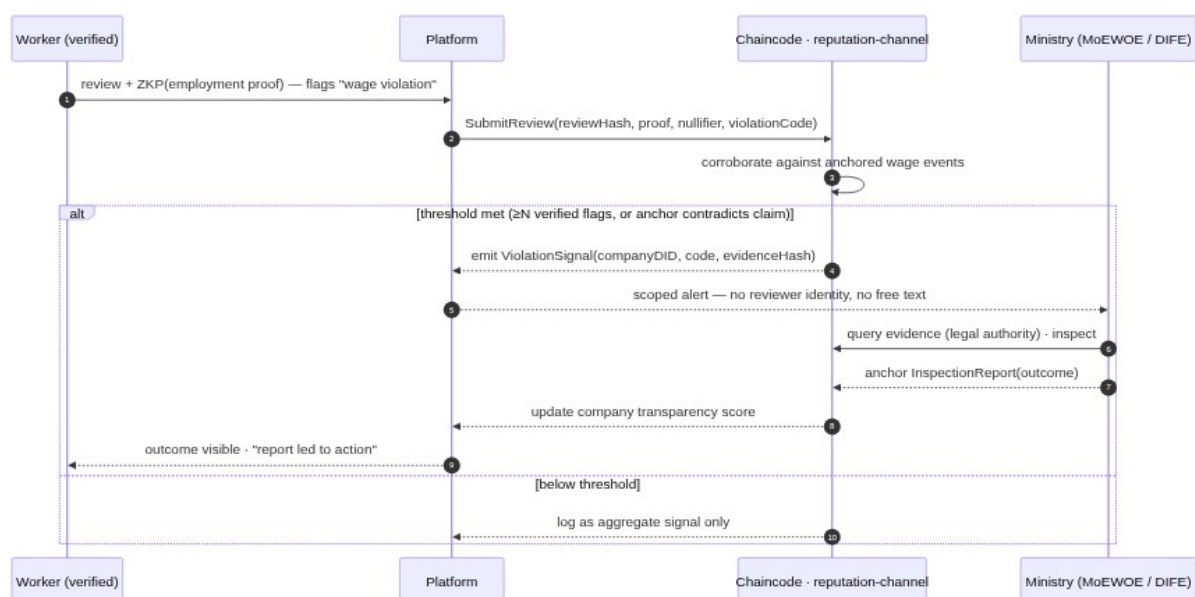


Figure 2 — The review-to-ministry enforcement loop

3.8 End-to-end journey

A worker earns a level-3 electrician skillCertificate at a TTC, signs an EmploymentContract co-endorsed by a licensed agency, and is issued an EmigrationClearance by BMET. A Dubai employer verifies her level and conduct in seconds against the public anchor — never contacting a Bangladeshi institution. Her monthly WageEvent, co-signed by the bank, quietly builds a verifiable income history. If a later employer fails to pay, she files a verified review flagging the wage violation; the corroborated signal reaches DIFE automatically, and the anchored InspectionReport records the outcome for every future worker. Across the whole journey, no single institution was trusted — and no single institution could have rewritten a single step.

4. Market and Partners

4.1 Market sizing

The serviceable market is defined by the four surfaces, each with an independent monetisable base:

Segment	Order of magnitude	Monetisable events
Regulator & agency tooling	~2,400 licensed agencies + the TTC network	Seat SaaS, per-clearance licensing
Employers (domestic + destination)	RMG and export-oriented firms; destination employers	Verification calls, Operations Suite seats
Corporate customers (Operations Suite)	Employers and agencies running managed workforces	Per-seat SaaS, per-deployment and per-clearance logistics fees
Financial sector	Banks, MFIs, insurers	Per income-verification call

As in the original design, a conservative initial market is a **single high-volume corridor**: even a low per-event fee over one corridor's clearance and verification volume funds a self-sustaining operation, and the network expands corridor by corridor rather than requiring national adoption on day one.

4.2 Partner ecosystem and their incentives

Partner	Role in the network	Their incentive to join
MoEWOE / BMET	Regulator; issues Identity Anchor and Emigration Clearance; enforcement responder	Fraud reduction, higher-skill export mix, a demonstrable worker-protection record, an automated violation early-warning system
RJSC	Company registry issuer	A registry that is actually trusted by banks, employers, and foreign partners
BFIU	AML/CFT observer of disclosures	Threshold-based ownership proofs instead of full, sensitive cap tables
BAIRA + licensed agencies	Industry node; co-sign contracts; operate the paid Suite	Faster placement, lower screening cost, a certified-ethical tier that commands better job orders
Employers (domestic + destination)	Issue employment proofs; consume verification; buy the Suite	A verifiable reputation that wins government tenders and foreign contracts
Banks / MFIs / insurers	Co-sign wage events; issue income verification	A new creditworthy segment with verified income
TTCs	Issue skill credentials	Differentiated, verifiable training quality
DIFE / labour courts	Enforcement verifiers and responders	Provable violations instead of testimony-vs-testimony
Foreign recruiters / employers	Verifiers, no network membership required	Self-contained proof with no local trust
Consortium foundation	Governance and network operations	A neutral, self-funding trust infrastructure

The **government's five roles** are worth stating explicitly, because they are what make the network governable without making it capturable: it is a **regulator-issuer** (clearance, registry), a ****node operator**, **an** oversight gatekeeper (**compliance veto, licensing, revocation**), a verifier and consumer (**inspection, tribunals, public procurement**), and a policy setter** in the governance channel — but its sovereignty is bounded to its own acts, and even its history is publicly anchored.

4.3 How partner incentives are allocated through the platform

Incentives are not promises in a partnership agreement; they are **encoded in the ledger**:

- **Verification fees** flow to the network per verification call, tiered by volume.
- **Seat subscriptions** from the Operations Suite fund the platform's common services.

- **Regulatory fees** attach to each clearance as a licence line.
- **Reputation stake** rises and falls with the reliability of each issuer's credentials — slashed on false attestation, earned on corroboration.
- **Corroboration scoring** rewards claims that independent parties can confirm.
- **The corporate trust signal** — a verified reputation — is itself the incentive that makes corporate customers pay, because it converts into government tenders and foreign job orders.

5. Competition and Risks

5.1 Direct competition

Competitor class	Examples	Why we differ
Professional networks	LinkedIn	Attested, not self-typed: skills and jobs are signed credentials, and endorsements are staked
Employer-review sites	Glassdoor, [Deshi Mula](https://deshimula.com/about-us)	Reviews are provably from real employees (verifiable anonymity), not anonymous-and-unverifiable
Blockchain credential networks	Velocity Network Foundation, Dock	Built for literate, white-collar Western markets; no migration clearance, contract substitution, wage, or enforcement coverage
National digital-locker schemes	Academic credential wallets	Education-only; employment, wage, conduct, and ownership are out of scope
HR / payroll SaaS	Workday, local payroll vendors	Their records live in the employer's database; ours are multi-party-signed and anchored
Generic SSI toolkits	Indy/Aries deployments	Primitives only; no governed network, no enforcement loop, no product

5.2 Indirect competition

The real incumbent is **manual trust**: apostilles, attestation, physical documents, and the paid middlemen who "verify" them. These processes are slow, expensive, and themselves the vector for the fraud the platform removes — the platform competes by making verification a local, seconds-long, tamper-evident operation that needs no middleman.

5.3 Business risks and mitigations

Risk	Mitigation
Government non-cooperation or regulatory change	Pilot with a single willing ministry; integrate as an <i>output</i> of existing processes rather than a replacement; align to e-government infrastructure
Two-sided cold start	Start from one corridor and one trade; the Operations Suite gives employers a reason to join before the worker side scales
Defamation / legal risk from reviews	Credential-gated reviews, no free-text rendering, anonymous reporters, anchored outcomes
UBO disclosure resistance	Threshold ZKP: prove compliance without publishing the cap table
Low-literacy adoption	Custodial mode, Shamir 2-of-3 recovery, Bengali-first wallet UX
Data-protection law (PDPA-style)	No PII on-chain; pairwise DIDs; selective disclosure by default
Paying-client capture (score inflation)	The score is derived from anchored evidence and publicly verifiable; the platform cannot edit it

5.4 Technical risks and mitigations

Risk	Mitigation
Key loss by workers	Shamir 2-of-3 social recovery; HSM-backed custodial mode with a self-custody path
Garbage in, gospel out	Corroboration scoring; issuer reputation staking; multi-party endorsement
Sybil issuers	MSP identity after regulator-verified existence; two-organisation onboarding endorsement; revocation path
Correlation / de-anonymization	Pairwise DIDs; per-employer link secrets; nullifiers unlinkable across employers
Zero-knowledge proof performance	Proofs verified off-chain at the edge; anchors on-chain are hashes, not payloads
Fabric single points (ordering, anchor)	Raft ordering across founding orgs; daily public-chain anchor
Regulator overreach / surveillance	Selective disclosure + legal-authority scoping; the platform cannot dump records on demand

6. Architecture

6.1 Platform selection

Employment Passport runs on **Hyperledger Fabric (permissioned)** with ****daily Merkle-root anchoring to a public chain****. The choice is deliberate and matches the trust problem:

- **Permissioned membership** — every participant holds an **MSP identity** issued only after regulator-verified legal existence. This is what makes "who attested this" meaningful: a review, a wage event, or a disclosure is signed by a *known, licensed institution*, not an anonymous public key.
- **Private data collections + channels** — sensitive operational data (full contracts, review text, UBO cap tables) is partitioned so each organisation holds only what it is entitled to, while all organisations hold the hash that proves integrity. This is the privacy model a public chain cannot offer natively.
- **Pluggable endorsement policies** — a write is valid only when the *required* parties co-sign it (e.g. employer + bank for a wage event). This encodes "no single party can fabricate a record" directly into consensus, which is the heart of the corruption argument.
- **Public anchoring** — the permissioned ledger alone still leaves "trust the consortium." The daily Merkle-root bridge to a public chain closes that gap: history becomes auditable by any third party holding a prior root.

The trade-off is stated openly: Fabric gives **governed multi-party trust**, not permissionless decentralisation. That is exactly what a regulated labour-and-corporate context requires, and the public anchor supplies the external accountability a public chain is good at.

System layers

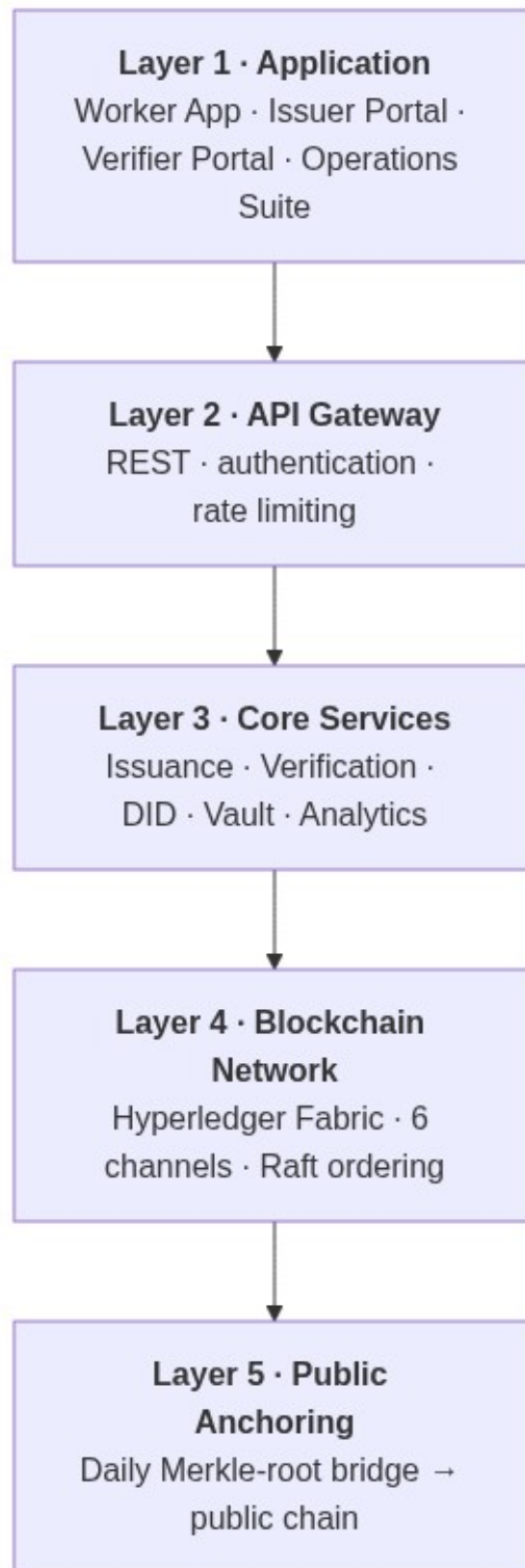


Figure 3 — The five-layer system architecture

6.2 Network topology and consensus

Founding consortium (6 organisations):

Node class	Members	Role
Labour regulator	MoEWOE / BMET	Identity anchor, emigration clearance, policy
Corporate regulator	RJSC	Company registry, UBO attestation
Financial intelligence	BFIU	Observer/verifier of disclosures, AML/CFT
Industry	BAIRA + licensed agencies	Contract co-signing, placement
Financial	Partner banks / MFIs / insurers	Wage-event anchors, credit, insurance
Foundation	Consortium foundation (non-profit)	Network operations, schema governance

Ordering & consensus: Raft (crash-fault tolerant) with one orderer per founding organisation — no single organisation controls ordering. **Endorsement policies are set per schema**, so the "quorum" is a *business* decision, not a global constant:

Schema	Endorsement policy	Rationale
EmploymentContract	employer + agency (2-of-2)	neither party can unilaterally invent a contract
WageEvent	employer + bank	a wage record needs the payer <i>and</i> an independent financial witness
EmigrationClearance	BMET	sovereign regulatory act
CompanyRegistration / UBO	company + auditor + RJSC	corporate identity is multi-party attested
WorkplaceReview	worker ZKP + nullifier check	see §6.5 — anonymity with one-voice
Endorsement	endorser + subject consent	attested, not self-typed

6.3 Channels and data segregation

Six channels partition visibility. Anything whose mere existence is sensitive lives in a **private data collection (PDC)** within its channel.

Channel	Members	Contents
anchor-channel	All organisations	Credential hashes, DID document pointers, revocation status lists, schema registry
recruitment-channel	BMET, BAIRA, agencies, employers	Job orders, clearance events, contract endorsements
payroll-channel	Banks, employers, BMET	Wage-event anchors, corroboration records
corporate-channel	RJSC, BFIU, BMET, banks, auditors, employers	Company registry, UBO threshold proofs, procurement awards, disclosures
reputation-channel	Employers, worker wallets, foundation	Review anchors (hash + proof), nullifiers, endorsement events, aggregate scores
governance-channel	Founding organisations only	Policy votes, chaincode upgrades, membership changes, fee parameters

6.4 On-chain versus off-chain

The rule from the original design is unchanged and now applied to the new surfaces: ****the ledger holds what must be provable; the vault holds what must be private.****

On-chain (ledger state)	Off-chain (encrypted vault)
Salted credential hash + schema identifier	Full credential JSON-LD document
Issuer DID + signature-suite identifier	Worker name, NID/passport, address, biometric refs
Review hash + review-proof commitment + nullifier	Full review text, sentiment
Aggregate score commitment (per company)	Individual rating breakdowns
UBO <i>threshold</i> proof (e.g. "no PEP ≥25%")	Full cap table, nominee details
Procurement award hash + conflict-check result	Tender documents
Revocation status list	—

6.5 Identity, credentials, and the nullifier registry

Identity stays pairwise: each worker/company uses a **pairwise DID** per verifier relationship, so presentations cannot be correlated across verifiers.

Credentials are issued with **BBS+ signatures**, preserving selective disclosure. New credential types extend the schema registry:

EmploymentProof · WorkplaceReview · Endorsement · CompanyRegistration · BeneficialOwnership · ProcurementAward · plus the existing EmploymentContract, EmigrationClearance, WageEvent, SkillCertificate.

The nullifier registry (new — the "one voice" mechanism for the workplace-transparency surface). A review must be provably written by a *real* current/former employee, yet the reviewer must stay anonymous *and* unable to post many reviews for the same employer. The mechanism:

1. A worker holds an EmploymentProof credential for employerDID.
2. To review, the wallet derives a **nullifier** $n = \text{PRF}(\text{linkSecret}, \text{employerDID})$ — a one-way value unique to (this worker, this employer) and unrevealing of either.
1. The wallet submits a **zero-knowledge proof** that (a) it holds a valid, unrevoked EmploymentProof for employerDID, and (b) n is correctly derived — **without revealing the credential, the worker, or the link secret**.
1. Chaincode appends n to the nullifier registry **if and only if it is absent**. A second review with the same n is rejected.

Result: **one verified voice per employee per employer**, with no linkage across a worker's different reviews (distinct link secrets) and no identity exposure to the platform or the company. This is the cryptographic upgrade that turns anonymous gossip into audit-grade transparency.

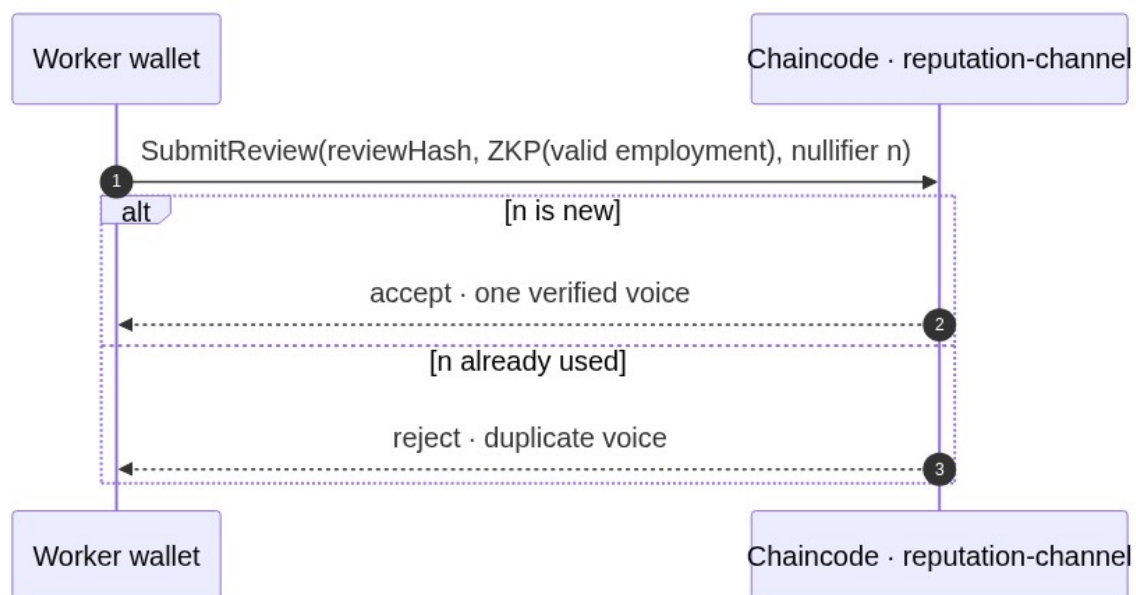


Figure 4 — The nullifier: one verified voice per employee

6.6 Key management

Unchanged, and now extended to the Operations Suite: **Shamir 2-of-3 social recovery** split across a BMET district office, a partner bank branch, and a family guardian; an optional HSM-backed custodial mode for low-literacy users; and a documented path to self-custody. Corporate keys are held in organisational HSMs with role-based quorum (no single employee can sign an attestation).

6.7 Integration with legacy systems

- **BMET emigration-clearance** — the clearance credential is emitted as a standard output of the existing national process (adapter, not replacement).
- **RJSC company registry** — the registry is *mirrored*, not rebuilt: RJSC remains authoritative and signs CompanyRegistration / UBO attestations; the ledger records the signed anchor.
- **Banks / MFIs / insurers** — wage events are integrated via a payroll adapter or core-banking file drop; the ledger receives the anchor, never the account data.
- **Foreign verifiers** — need *no* integration: they verify a self-contained BBS+ proof plus the public-chain Merkle proof. No Fabric membership, no API key, no "trust the ministry."

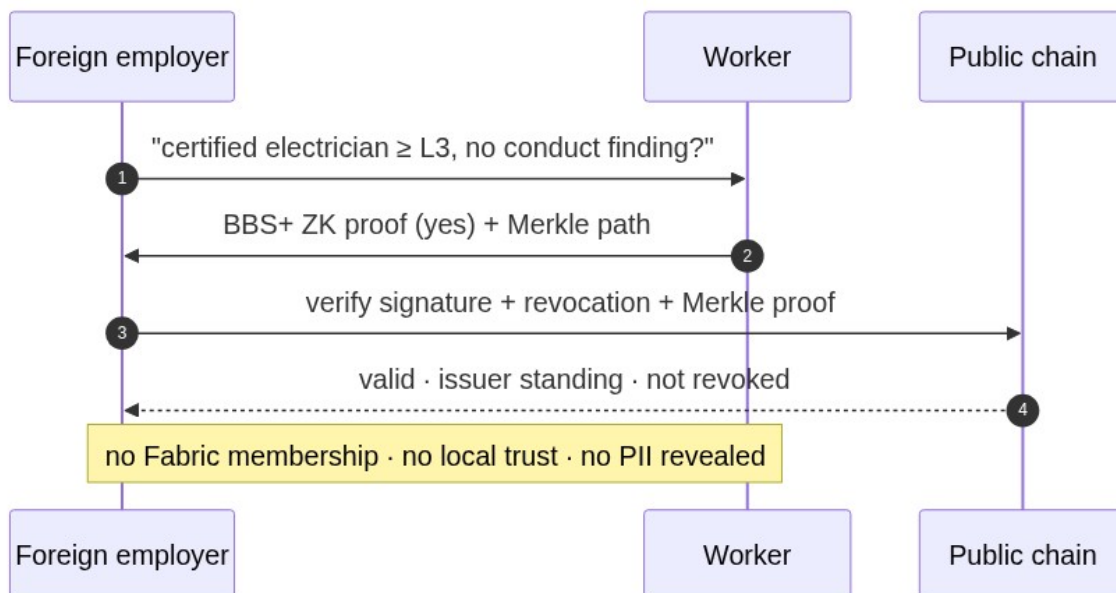


Figure 5 — Foreign verification without local trust

6.8 Public anchoring and the transparency digest

The daily **Merkle-root bridge** is retained, with a second, purpose-built output: the **transparency digest**. For each company the platform publishes a verifiable summary —

"12 verified reviewers · 74% recommend · 3 wage events reconciled this month · UBO thresholds satisfied · no open conduct findings."

Each figure is a commitment whose root is in the anchored Merkle tree, so the public can verify the aggregate is computed over the real evidence **without seeing any raw review, salary, or owner**. This is what makes the *platform itself* unable to quietly inflate or suppress a company's score — the corruption defence applied to the transparency layer.

6.9 The Corporate Operations Suite (application layer)

The paid Operations Suite is an **application-layer client, not a privileged write path**. Its modules (onboarding + bulk issuance, payroll anchoring, clearance/visa tracking, deployment logistics, benefits, offboarding, analytics) all call the **same chaincode under the same endorsement policies** as any other participant.

This is the key design point: a corporate customer pays for *workflow, tooling, and logistics* — not for a backdoor. The Suite cannot mint a wage event without the bank's endorsement, cannot issue a clearance without BMET, and cannot delete a review. Running payroll and clearance through the Suite therefore *is* the transparency surface: the same act that manages the workforce produces the reconciled, verifiable evidence the platform publishes.

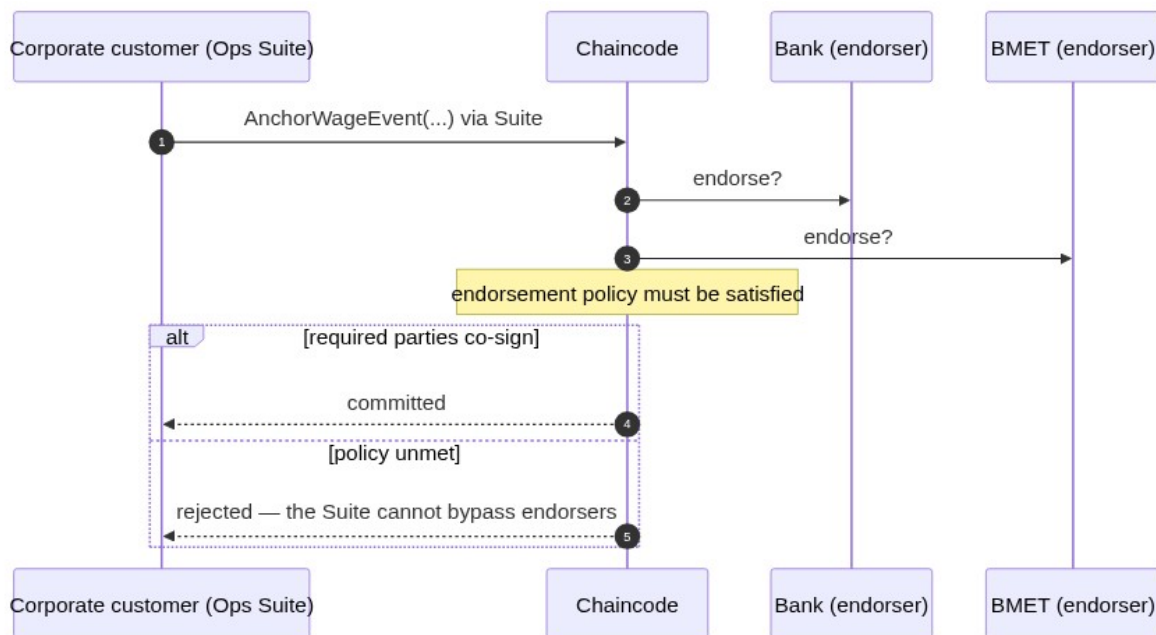


Figure 6 — The Operations Suite as an unprivileged client

6.10 Performance and capacity

Anchors are tiny (hashes + proofs), and verification is an off-chain ZK proof check, so the performance-critical paths are cheap: a review submission is one nullifier existence check; a foreign verification is a local proof check + a Merkle-path lookup. The ledger's job is **integrity, not throughput**; documents and payloads never touch it.

7. Governance

7.1 Network membership governance

- Organisations join by submitting legal documents and licences and signing the network charter; existing members vote on admission with a **regulator compliance veto**.
- Members are removable on licence loss or misconduct; ****previously issued credentials remain valid and verifiable**** even after an organisation leaves.
- Roles are Issuer, Verifier, Regulator, and Operator with distinct permissions; new member classes (RJSC, BFIU, auditors, foreign verifiers) inherit the same on/off-boarding discipline.
- Key rotation, incident response, and security audits are mandatory and scheduled.

7.2 Business network governance

- A **non-profit consortium foundation** governs rules, schemas, and common services; its board includes regulators, industry, banks, training organisations, and a **worker representative**.
- No single organisation holds full control; disputes use evidence collection → mediation → competent authority, with the platform supplying evidence but never deciding the case.

7.3 Technology infrastructure governance

- Each organisation operates its own infrastructure; government nodes stay in government facilities.
- Technology updates require security review, testing, and member approval; ****new on-chain fields pass a privacy review first****.
- A **4-of-5 organisation agreement** is required to pause writes in a major emergency.
- The Operations Suite ships under a published **SLA and compliance schedule**, audited against the same governance rules as any network service.

7.4 Asset tokenisation model

The platform tokenises **four non-transferable asset classes** — no cryptocurrency, no tradable network token, all fees settled in fiat (BDT) through regulated rails:

Asset class	Unit	Transferable?	Economic purpose
Credential	one per claim, bound to subject DID	No	Prevent a credential market (the core fraud vector)
Reputation stake	issuer standing	No — slashable, not sellable	Align issuers: good attestations build standing; false ones are slashed
Review-vote weight	one per (worker, employer), spent as a nullifier	No	One verified voice per employer — the anti-astrourfing unit
Logistics work-order	one per deployment/clearance task, lifecycle-stateful	No	Auditable unit of the Operations Suite; proves each step happened

The **non-transferability is the security property, not a limitation**: transferability would create a market for credentials, reputation, and votes — precisely the corruption the platform exists to remove. The public-chain anchoring fees are paid in fiat through a regulated service provider, so no participant holds or is exposed to a volatile digital asset.

8. Valuation and Distribution

8.1 Value created

Beneficiary	Value
Worker	Lower recruitment cost through disintermediation of unverifiable middlemen; wage premium for provable skill; admissible evidence in wage and substitution disputes; access to formal credit on verified income
Domestic employer / agency	Screening cost and time-to-hire collapse from weeks to seconds; a credible quality signal that lets compliant firms escape price competition
Foreign employer	Confident hiring of verified skills with no local trust required; lower mis-hire and fraud
Corporate customer (Operations Suite)	Managed, compliant workforce; a verifiable reputation that wins government tenders and foreign contracts
Government (MoEWOE/BMET, RJSC, DIFE)	Fraud reduction, a higher-skill export mix, enforceable labour law, verifiable companies, and an automated violation early-warning system
Financial sector	A new creditworthy segment with verified income; lower default risk on migrant lending
National economy	Skill upgrading raises remittance per worker — the highest-leverage variable in the sector

8.2 Revenue model

Stream	Payer	Basis
Corporate Operations Suite	Employers, agencies	Per-seat SaaS + per-deployment + per-clearance logistics fees — the primary stream
Verification-as-a-service	Employers, agencies, verifiers	Per verification, tiered by volume
Financial-services API	Banks, MFIs, insurers	Per income-verification call
Direct-hire marketplace	Employers	Success fee on placements made without agency intermediation
Aggregate analytics	Government, development agencies, researchers	Subscription to corridor and compliance intelligence
Regulatory licensing	Government	Per-clearance licence fee bundled into the existing emigration-clearance process

The **premium incentive** is the flywheel that makes the model self-reinforcing: a corporate customer pays for the Operations Suite because it converts into **contract value** — verifiable reliability to foreigners and government → more tenders, job orders, and subcontracts won → more revenue → more reason to pay. The trust is *earned* from anchored evidence, never *purchased*: there is no backdoor and no way to buy the score, which is precisely why the signal is worth anything. The ministry early-warning dashboard is delivered as a **public good**, not a revenue line.

8.3 Go-to-market and roadmap

Phase	Window	Objective	Success measure
0 — Prototype	Months 0–6	Fabric network with three organisations; wallet and verifier SDK; one corridor, one trade; review prototype	Working end-to-end issuance, ZK verification, and review submission; BCOLBD prototype round
1 — Corridor pilot	Months 6–18	Two licensed agencies, one TTC, one destination employer; review surface beta; Operations Suite MVP	5,000 workers; measured reduction in time-to-verify and screening cost
2 — Ops Suite GA + regulator	Months 18–30	BMET clearance live; bank wage events; ministry trigger live	100,000+ workers; clearance credential issued as standard output
3 — Destination + corporate	Months 30–48	Bilateral recognition in one major corridor; RJSC/UBO registry; procurement transparency	Employer-side verification volume exceeds issuance — the network-effect inflection
4 — Regional	Year 4+	Cross-border W3C schema interoperability across South Asia	Credential recognition; self-funding foundation

9. Conclusion

Employment Passport began as a way to make a worker's credentials travel. It is now a corporate-transparency platform, because the same root cause — ****institutions and paperwork that cannot be trusted**** — corrupts every layer of Bangladesh's labour market at once: a worker's history, a company's identity, a company's reputation, and the enforcement of the laws that protect workers.

A conventional system would have to pick a single administrator to trust, and in this environment that choice is exactly what fails. The platform therefore does something a database cannot: it spreads control across the institutions that distrust each other, signs every record with the parties responsible for it, and anchors the whole history to a public chain so that even the platform cannot rewrite its own past. The result is not a promise of honesty — it is a system in which honesty is no longer required for trust, because every claim can be verified against a proof that no single party controls.

That is the difference between "trust us" and "verify the proof" — and it is the difference between a labour market that runs on forged paper and one that runs on provable facts.

Appendix A — Core Chaincode Interface

Function	Caller	Effect
RegisterDID(did, docHash)	Issuer (BMET)	Registers a DID document pointer on anchor-channel
IssueCredential(credHash, schemaId, issuerDID, subjectDID, expiry)	Authorised issuer	Anchors a credential; endorsement policy enforced per schema
RevokeCredential(credHash, reasonCode)	Original issuer or regulator	Sets the revocation bit in the status list
VerifyAnchor(credHash)	Any verifier	Returns issuance state, issuer standing, endorsement set, corroboration score
RecordDisclosure(consentHash, verifierDID)	Worker wallet	Logs a consent event so the worker can audit who checked what
SubmitCorroboration(credHash, sourcedID, evidenceHash)	Secondary issuer	Raises the corroboration score of an existing claim
SubmitReview(reviewHash, proof, nullifier, violationCode)	Worker wallet	Accepts a review only if the employment proof is valid and the nullifier is unused
EmitViolationSignal(companyDID, code, evidenceHash)	Chaincode (trigger)	Fires when a corroborated violation threshold is met; routes to the ministry
RegisterCompany(companyDID, docHash)	RJSC	Registers a company DID and registry anchor on corporate-channel
ProveUBO(companyDID, thresholdProof)	Company + auditor + RJSC	Anchors a beneficial-ownership threshold proof
AnchorProcurement(awardHash, conflictCheckHash)	Procuring entity	Anchors a tender outcome and conflict-check result
UpdateAgencyStanding(agencyDID, delta, evidenceHash)	Regulator + observer	Adjusts reputation stake; dual endorsement required
ProposeSchema(schemaDoc) / VoteSchema(id, vote)	Founding organisations	Governance-channel schema lifecycle

Appendix B — Example Credential Schemas (abridged)

Fields marked **are** disclosable independently under BBS+; fields marked **are** held only off-chain.

WorkplaceReview

Field	Type	Notes
subjectDID	DID	Pairwise, per verifier
companyDID	DID	The employer being reviewed
reviewHash	hash	Salted hash of the review content
nullifier	scalar	One-time PRF(LinkSecret, companyDID); prevents duplicate voices
violationCode	enum	Optional wage/contract/overtime/safety/termination flag
reviewText	encrypted blob ref	Never on-chain; stored in the off-chain vault

BeneficialOwnership

Field	Type	Notes
companyDID	DID	
thresholdProof	ZKP	Proves "no owner above X% is sanctioned/PEP" without the cap table
attestationDate	date	
capTable	encrypted blob ref	Full ownership structure, off-chain only

Appendix C — Glossary

- **Verifiable credential** — a signed digital statement about a subject that can be selectively disclosed and independently verified.
- **BBS+ signature** — a signature scheme supporting selective disclosure and zero-knowledge

predicates over signed attributes.

- **DID / pairwise DID** — a decentralised identifier; *pairwise* means a distinct DID per verifier relationship to prevent correlation.
- **Zero-knowledge proof (ZKP)** — a proof that a statement is true without revealing why.
- **Nullifier** — a one-time value ($\text{PRF}(\text{LinkSecret}, \text{companyDID})$) that proves a unique voice without revealing identity.
- **Link secret** — a private per-employer secret used to derive nullifiers and prove employment.
- **Endorsement policy** — the rule defining which parties must co-sign a write for it to be valid.
- **Channel / private data collection** — Fabric constructs that partition visibility across members.
- **MSP** — Membership Service Provider; the certificate authority that issues network identities.
- **UBO** — ultimate beneficial owner.
- **Transparency digest** — a public, Merkle-committed aggregate (counts, scores, outcomes) verifiable without raw data.
- **Corroboration score** — the confidence weight of a claim based on how many independent parties confirm it.

Appendix D — Data Sources

Figures cited in §2 and §4 are drawn from public sources and should be re-verified against the primary source before final submission, as migration statistics are revised frequently.

1. Bureau of Manpower, Employment and Training (BMET), annual overseas employment statistics, 2025 — total departures, destination distribution, skill-category breakdown.
1. The Business Standard / Prothom Alo reporting on BMET data, 2025–2026 — year-on-year overseas employment growth, remittance inflows, fiscal-year departure figures.
1. Bangladesh Bureau of Statistics, Survey on the Use of Remittances — average migration cost and cost-recovery period.
1. World Bank / KNOMAD, Migration and Development Brief — comparative worker-paid recruitment costs.
2. International Organization for Migration, Bangladesh Household Remittance Survey — distribution of migration costs between licensed agencies and intermediaries.
1. Deshi Mula, "About Us" — Bangladesh IT graduate supply vs. demand figures ($\approx 20,000$ CSE graduates vs. 7,500–8,000 openings), cited for the employer-review market context.